Kaiming Liu

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Research Profile

Machine learning for probabilistic inference and robust prediction in scientific inverse problems with multiple plausible solutions.

Education

University of Illinois Urbana-Champaign
Ph.D. in Physics, in progress Aug 2023–present
Advisor: Bryan K. Clark

Xi'an Jiaotong University
B.S. in Physics (Honors) Sep 2019–Jul 2023
Special Class for the Gifted Young Sep 2017–Jul 2019

University of California, Berkeley
BPIE Program (non-degree visiting program) Aug–Dec 2021

Current Research

University of Illinois Urbana-Champaign
Graduate researcher, Bryan K. Clark group Nov 2025–present

Conditional density modeling / crystal-field parameters

- Built residual-MLP conditional Gaussian mixtures for conditional density estimation from magnetic susceptibility.
- Represented multiple plausible solutions and checked them by reconstructing susceptibility; studied robustness on simulated and experimental data.
- Evaluated conditional Gaussian mixtures and conditional RealNVP normalizing flows in a pilot density-model comparison.

Robust inverse prediction / neutron scattering

- Trained MLP and ResNet conditional Gaussian mixtures to infer exchange parameters from high-dimensional neutron-scattering maps.
- Checked predictions with Sunny forward simulations and classical fit references; tested noise, intensity scaling, and background offsets.

Neural-network backflow / JAX and Monte Carlo

- Implemented neural-network backflow and variational Monte Carlo in JAX for Hubbard models, with GPU/multi-GPU sampling.
- Profiled local-energy evaluation and gradient synchronization; verified chunked evaluation and global reductions against reference computations.

Publications

[P1] **Evidence of intertwined pair density and charge density wave orders in UTe_2**
Zhen Zhu, Yudi Huang, Julian May-Mann, **Kaiming Liu**, Zheyu Wu, Shanta R. Saha, Johnpierre Paglione, Alexander G. Eaton, Andrej Cabala, Michal Vališka, Eduardo Fradkin, Vidya Madhavan.
PNAS 123 (29), e2602117123 (2026). arXiv.

Prior Research

University of Illinois Urbana-Champaign

Research assistant, Vidya Madhavan group — Aug 2023–Oct 2025

- Grew thin-film heterostructures with MBE and characterized electronic structure with STM/STS in ultrahigh vacuum.
- Automated growth temperature control and built Python/MATLAB tools for quasiparticle-interference analysis.

Xi'an Jiaotong University

Research assistant, Yongchang Zhang group — Mar 2022–Jul 2023

- Simulated Rydberg-atom dynamics and photon-mediated interactions in Python/MATLAB to study quantum nonlinear optics.
- Built a discrete-time lattice model of two-qubit phase-flip dynamics and many-body regimes.

University of California, Berkeley

Research assistant, Michael F. Crommie group — Oct–Dec 2021

- Assisted synthesis and STM characterization of graphene nanoribbons with five-membered rings to study their electronic states.

Teaching

University of Illinois Urbana-Champaign

Teaching assistant, PHYS 212 — Spring & Fall 2026
Teaching assistant, PHYS 102 — Aug–Dec 2023

Selected Honors

China National Scholarship — 2021
Xi'an Jiaotong University

Mathematical Contest in Modeling, Meritorious Winner — 2021
COMAP

Technical Skills

ML / modeling: PyTorch, JAX; conditional mixtures, residual networks, probabilistic inference.
Research engineering: Python, NumPy, SciPy, Git, Linux; GPU/multi-GPU and cluster computation, runtime profiling.
Numerical / scientific methods: MCMC, VMC, numerical optimization; forward-model evaluation and simulation-to-experiment robustness.
Experimental physics: MBE, STM/STS, UHV; feedback control, automation, MATLAB and data analysis.